

■ CoefficientList

`CoefficientList[poly, var]` gives a list of coefficients of powers of *var* in *poly*, starting with power 0.

`CoefficientList[poly, {var1, var2, ...}]` gives a matrix of coefficients of the *var_i*.

Example: `CoefficientList[x^2 + 2 x y - y, {x, y}]` $\longrightarrow$ `{{0, -1}, {0, 2}, {1, 0}}`. ■ The dimensions of the matrix returned by `CoefficientList` are determined by the values of the `Exponent[poly, vari]`. ■ Terms that do not contain positive integer powers of a particular variable are included in the first element of the list for that variable.

■ `CoefficientList` always returns a rectangular matrix. Combinations of powers that do not appear in *poly* give zeroes in the matrix. ■ See page 383. ■ See also: `Series`, `Collect`, `FactorList`.